Gamma positivity of the diagonal numerators of the blade array

Principia Math In Progress

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Task 1: Complete solution of MathDB 340676

The target is Conjecture 8.4 of Early's current preprint [1]. Its symmetry and coefficient-sum assertions were already proved by Donghyun Kim in Appendix A of that paper. The remaining unimodality assertion follows from the stronger gamma positivity proved below. We use Early's established enumeration formula, Proposition 8.3:

$$T_{n,k} = \sum_{i=1}^{n} S(n,i)c(i-1,k-1) \qquad (1 \le k \le n),$$

where S and c are the Stirling numbers of the second and unsigned first kinds. To fix the diagonal indexing precisely, put

$$H_j(x) = \sum_{r\ge 0} T_{j+r+1,r+1}x^r, \qquad P_j(x) = (1-x)^{2j+1}H_j(x), \quad j \ge 0.$$

This gives the source's numerators 1, $1+x$, $1+10x+x^2$, and so on.

Theorem 1. *For every $j \ge 0$ there are nonnegative integers $q_{j,r}$ such that*

$$P_j(x) = \sum_{r=0}^{\lfloor j/2 \rfloor} 4^r q_{j,r} x^r (1+x)^{j-2r}, \qquad q_{j,0} = 1. \tag{1}$$

Consequently P_j has symmetric, unimodal, nonnegative integer coefficients, and

$$P_j(1) = \frac{(2j)!}{j!}.$$

Its coefficients strictly increase up to the middle, apart from the necessary pair of equal central coefficients when j is odd.

All series manipulations in the proof are formal, over fields of characteristic zero; no analytic convergence is required.

The row series and a formal inverse

Set

$$R_m(y) = \sum_{k=0}^{m} T_{m+1,k+1}y^k, \qquad G(z) = -\log(2-e^z).$$

The established row exponential generating function is

$$\sum_{m\geq 0} R_m(y)\frac{z^m}{m!} = e^z(2-e^z)^{-y} = e^z e^{yG(z)}. \tag{2}$$

This formula is also recorded by Skirrow in OEIS A320280 [2]. For completeness, it follows directly from Early's formula: since $\sum_k c(i-1,k)y^k = y^{\overline{i-1}}$ and $\sum_{n\geq i} S(n,i)z^n/n! = (e^z-1)^i/i!$, differentiation in z gives

$$\sum_{m\geq 0} R_m(y)\frac{z^m}{m!} = e^z \sum_{i\geq 1} y^{\overline{i-1}}\frac{(e^z-1)^{i-1}}{(i-1)!}.$$

The binomial theorem proves (2). In particular,

$$T_{j+r+1,r+1} = \frac{(j+r)!}{r!}[z^{j+r}]e^z G(z)^r. \tag{3}$$

Let $w = w(u,x)$ be the unique series with $w(0,x) = 0$ satisfying

$$u = w - xG(w) = w + x\log(2-e^w). \tag{4}$$

The linear coefficient $1-x$ is invertible in $\mathbb{Q}[[x]]$, so this inverse exists as a formal series in u. We claim that

$$H_j(x) = j![u^j]\frac{d}{du}e^{w(u,x)}. \tag{5}$$

Indeed, changing variables in a formal residue and expanding in x gives

$$\begin{aligned} j![u^j]\frac{d}{du}e^{w(u,x)} &= j!\,\mathrm{Res}_{z=0}\,\frac{e^z}{(z-xG(z))^{j+1}}\,dz \\ &= \sum_{r\geq 0}\frac{(j+r)!}{r!}x^r[z^{j+r}]e^z G(z)^r = H_j(x). \end{aligned}$$

The formal change of variables is valid for an invertible series with zero constant term. Expansion in x makes each displayed residue coefficient well defined, and the last equality is (3).

A differential equation with nonnegative coefficients

Write $A = 1+x$, $B = 1-x$, and $V = e^{w(u,x)}$. Differentiating (4), with $V(0,x) = 1$, yields

$$\frac{dV}{du} = \frac{V(2-V)}{2-(1+x)V}.$$

Introduce a new variable s and a parameter h by

$$s = \frac{A}{B^2}u, \qquad V = 1 + \frac{B}{A}Y, \qquad h = \frac{4x}{A^2}.$$

Since $A^2 - B^2 = 4x$, the resulting initial-value problem is

$$\frac{dY}{ds} = f_h(Y), \quad Y(0,h) = 0, \qquad f_h(y) = 1 + y + \frac{hy^2}{1-y}. \tag{6}$$

Define, first with h an independent indeterminate,

$$Q_j(h) = \left(f_h(y) \frac{d}{dy} \right)^j f_h(y) \Bigg|_{y=0}.$$

Repeated differentiation of (6) shows that $Q_j(h) = \partial_s^{j+1} Y(0,h)$. Since $V_u = B^{-1} Y_s$ and $s_u = A/B^2$, equation (5) becomes

$$H_j(x) = \frac{A^j}{B^{2j+1}} Q_j\left(\frac{4x}{A^2} \right). \tag{7}$$

Lemma 1. *$Q_j(h) \in \mathbb{Z}_{\geq 0}[h]$, $\deg Q_j \leq \lfloor j/2 \rfloor$, and $Q_j(0) = 1$.*

Proof. Expand $f_h(y) = 1 + y + h \sum_{a \geq 2} y^a$. Multiplication by f_h and differentiation in y preserve nonnegative integer coefficients. At each fixed j only finitely many terms contribute at $y = 0$.

For the degree bound, assign the monomial $h^r y^a$ weight $a - 2r$. Every monomial of f_h has weight at least zero. Differentiation lowers weight by one, and multiplication by f_h cannot lower it. Thus every monomial of $(f_h \, d/dy)^j f_h$ has weight at least $-j$. After evaluation at $y = 0$, a surviving h^r must satisfy $-2r \geq -j$. Finally, at $h = 0$, the operator is $(1 + y)d/dy$ and fixes $1 + y$, giving $Q_j(0) = 1$. □

Writing $Q_j(h) = \sum_r q_{j,r} h^r$ in (7) and multiplying by B^{2j+1} proves (1). Each polynomial $x^r(1+x)^{j-2r}$ is symmetric about $j/2$ and unimodal when its coefficient vector is padded with zeros to length $j + 1$. Nonnegative sums preserve these two properties because all summands have the same center. The $r = 0$ summand $(1+x)^j$ has coefficient one and gives strict growth to the middle. This proves the new unimodality assertion.

For an additional derivation of the already established coefficient sum, specialize (6) at $h = 1$. Then $Y_s = 1/(1 - Y)$, so $Y = 1 - \sqrt{1 - 2s}$ and $Y_s = (1 - 2s)^{-1/2}$. Therefore

$$Q_j(1) = (2j - 1)!!, \qquad P_j(1) = 2^j Q_j(1) = \frac{(2j)!}{j!},$$

including $j = 0$ with $(-1)!! = 1$. This completes the theorem.

Examples, verification, and scope

The first polynomials Q_j are

$$1, \quad 1, \quad 1 + 2h, \quad 1 + 14h, \quad 1 + 88h + 16h^2, \quad 1 + 628h + 316h^2.$$

The accompanying exact verifier compares the original Stirling sums with the gamma formula, checks a separate row recurrence, solves the cleared differential equation independently at six integer parameters, and checks the source tables and coefficient properties. These finite checks support reproducibility; the proof above treats every diagonal.

Early's enumeration formula, Kim's symmetry and factorial-sum proofs, and Skirrow's recorded row generating function are prior inputs and are not claimed as new. The contribution here is the positive differential-operator formula for the diagonal gamma coefficients and the resulting all-diagonal unimodality. Review is by the authoring agent only, without independent peer review or formal proof-assistant certification.

Task 2: A combinatorial interpretation of the gamma coefficients

Find a direct statistic on the canonical blade objects whose distribution gives $q_{j,r}$, or explain the factor 4^r by a decoration of those objects. Equation (6) suggests increasing trees with r vertices having at least two children, but a correspondence with blades is still needed.

Task 3: Zeros and interlacing

Determine whether every $Q_j(h)$ has only real negative zeros, and whether the nonconstant successive polynomials satisfy an interlacing relation. The proof establishes coefficient positivity, not these stronger properties.

Task 4: A limiting coefficient distribution

For $\Pr(X_j = a) = [x^a]P_j(x)/P_j(1)$, symmetry gives $\mathbb{E}X_j = j/2$. Determine the variance asymptotics and whether the centered, normalized variables satisfy a central limit theorem.

Task 5: Refine the branching degrees

Replace $hy^2/(1-y)$ by $\sum_{a\geq 2} h_a y^a$ in (6). Interpret the resulting nonnegative multivariate coefficients and identify a refinement of the blade enumeration that specializes back to P_j. These four followups are questions, not additional solved MathDB problems.

References

[1] N. Early, with an appendix by D. Kim, *Honeycomb Tessellations and Graded Permutohedral Blades*, arXiv:1810.03246v4, 3 February 2025, Proposition 8.3, Conjecture 8.4, and Appendix A.

[2] OEIS Foundation, A320280, row generating functions recorded by N. L. Skirrow, 22 December 2025.